

Path 1: Pre Security > Module 5: Operating Systems Basics > Room 1: Operating Systems: Introduction

<https://tryhackme.com/room/operatingsystemsintroduction>

>> Task 1: Introduction

♦ Scenario:

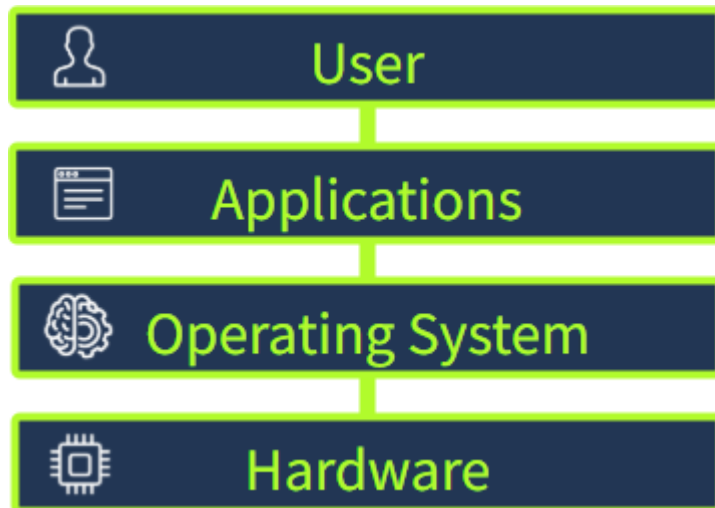
Your friend just upgraded their primary computer and, knowing you're getting into tech, has generously gifted you their old machine. It's been sitting untouched for a while, and he can't remember much about it, only that it "used to run well". Before you can decide whether to upgrade it, wipe it, sell it, or turn it into your next project, you need to figure out precisely what you're dealing with.

♦ Learning Objectives:

- Understand what an operating system is and the role it plays
- Explain the core duties of an operating system
- Identify common OS types and their typical use cases
- Practice interacting with an OS to gather system information

>> Task 2: The Invisible Manager

An **operating system** (OS) is the core software that coordinates everything happening on a computer. It sits between the user, applications, and the system's physical hardware, acting as the invisible manager that keeps the entire machine running as one unified system.



A helpful analogy is to think of your computer as a busy airport, with all its components functioning together.

- Your **hardware** (CPU, RAM, storage, connected devices): The runways, airplanes, fuel systems, radar, and other physical infrastructure.
- Your **applications** (web browser, game launcher): The various airlines and their passengers, all trying to take off, land, and request services.
- Your **operating system** (Windows, Linux, macOS): The entire air traffic control system, directing all of this activity. It schedules resources, manages traffic, resolves conflicts, and ensures safety.

- ◆ System Privilege Layers:

Inside a modern computer, different parts of the system operate at various permission levels. Some components can communicate directly with the hardware, while regular applications run in a safer, restricted environment. This separation is intentional and helps prevent conflicts and security issues.

- **Kernel space:** The privileged, locked-down core of the OS. This is where the kernel, the part of the operating system that directly manages hardware and system resources, runs. It has unrestricted access to the CPU, memory, storage, and all hardware components.

- **User space:** Where all standard applications run. Applications in the user space are deliberately prevented from accessing hardware directly. Whenever they need to open or save a file, play a sound, or connect to Wi-Fi, they must make a system call and request that the kernel act on their behalf.

- ◆ Operating System Duties:

Now that you know what an operating system is and how system privilege is separated, let's look at what it actually does behind the scenes. Every OS is responsible for a few core duties that allow your computer to run safely, efficiently, and predictably.

OS Responsibility	What the OS Does	Example
Process Management	Creates, schedules, prioritizes, and terminates running programs. The <u>OS</u> decides how much <u>CPU</u> time each process gets, making multitasking feel seamless	Opening multiple apps, like your browser, music player, and social media, without your computer freezing
Memory Management	Allocates <u>RAM</u> to processes, protects the app's memory from other processes, and reclaims memory when apps are closed. When <u>RAM</u> runs low, the <u>OS</u> uses virtual memory to keep your system stable	Opening multiple app at once, the <u>OS</u> allocates <u>RAM</u> to each one and keeps them isolated so they don't interfere or crash each other
File System Management	Organizes files into directories, handles	Creating a new folder, saving a photo, or setting

	naming, paths, permissions, metadata (name, size, type, timestamps)	a file to "read only"
User Management	Handles multiple user accounts, authentication, and permissions to determine who can access what	Logging in with your password and keeping your files inaccessible to other user accounts
Device Management	Loads drivers and provides a universal interface (hardware abstraction layer), so apps can say "print this" or "play this sound"	Plugging in a new mouse, printer, or external hard drive and having it work immediately

♦ Operating System Security:

It is important to understand that every OS also acts as a security foundation. Before any antivirus, firewall, or security tool is introduced, the OS is already enforcing protections in the background, some of which we covered above.

At a basic level, your operating system handles

- **Authentication:** Verifies who you are through login passwords and biometrics
- **Permissions:** Controls exactly what each user and app is allowed to read, write, or execute
- **Isolation:** Keeps every process in its own protected box (kernel/user space separation)
- **System Protection:** Safeguards critical system files and settings from unauthorized changes

Q1) Which OS space has unrestricted access to your computer's hardware?

Answer: Kernel Space

Q2) Which OS responsibility manages user accounts, authentication, and permissions?

Answer: User Management

>> Task 3: OS Interaction and Landscape

♦ OS Interfaces:

Now that you have a solid understanding of the operating system and its various responsibilities, let's look at how we interact with the OS. Interaction with the OS can be divided into two main parts: the **graphical user interface** (GUI) and the **command-line interface** (CLI).

♦ The Operating System Landscape:

Operating System Type	Primary Use Case	Key Characteristics
Desktop	Personal computers, daily work, gaming, content creation	Rich graphic interface, runs many apps at once, user-focused
Server	Web hosting, databases, cloud services, back-end	Headless (no GUI), maximum uptime, multi-user, remote access
Mobile	Smartphones and tablets	Touch-based UI, power efficient, always connected, app sandboxing
Embedded	Appliances, cars, IoT devices, smart TVs, routers	Tiny footprint, runs on limited hardware
Virtual/Cloud	Virtual machines, containers, cloud instances	Lightweight, scalable, rapid deployment

♦ Real World Operating Systems:

Desktop:

- **Windows**: The most widely used operating system on personal computers.

Windows 10 (end-of-life), Windows 11

- **macOS**: Apple's desktop OS, known for its polished GUI and integration with other Apple devices.

Sonoma (14), Sequoia (15), Tahoe (26)

- **Linux:** Not a single OS but a family of open-source operating systems called distributions.

Ubuntu, Debian, Fedora

Server:

- **Windows:** Used in large networks, data centers, and corporate environments.
Server 2016, 2019, 2022, 2025

- **Linux:** The vast majority of web servers, trusted for its reliability and open-source nature.

Ubuntu Server, Debian, CentOS, Red Hat

- **Unix:** Large enterprises, finance, telecom, government.
IBM AIX, Oracle Solaris

Mobile:

- **Android:** The most widely used mobile OS, which runs on phones, tablets, and smart devices.

Android 14 - 16, Manufacturer versions

- **iOS:** Apple's mobile OS running on iPhones, iPads, and other devices.
iOS 17, 18, 26

Embedded and IoT Devices:

- **Embedded Linux:** Specialized OS built into devices with dedicated functions.
OpenWrt, Ubuntu Core, Yocto Project

- **Real-Time OS:** Designed for apps where tasks need guaranteed response times (aircraft controls).

FreeRTOS, VxWorks, QNX

Virtual and Cloud:

- **Cloud/VM:** Massive data centers that host websites, apps, and streaming services
Ubuntu LTS, Amazon Linux, Rocky Linux

- **Container-optimized:** Lightweight alternatives to VMs that package just the app and its dependencies
Alpine Linux, Bottlerocket AWS, Flatcar Linux

- ♦ Why So Many Operating Systems?

Different devices and environments require different capabilities from an OS. A laptop must be user-friendly and support multitasking. Servers require stability, security, and must be able to run continuously without interruption. Mobile devices need power efficiency and hardware integration to extend battery life. Embedded systems use lightweight operating systems designed for a specialized purpose.

The companies and communities that develop these operating systems also have their own goals. Some focus on ease of use, performance, security, openness, or customization. Because each environment values different capabilities, no single OS is the perfect fit for every situation. Instead, an ecosystem of operating systems has evolved.

>> Task 4: Conclusion

- ◆ Key Terminology:

Operating system (OS)

The core software that manages hardware, applications, and all system resources.

Kernel space

The OS's highly privileged area with direct hardware access, and the home of the kernel, which directly manages hardware and system resources.

User space

The area where regular applications run with limited permissions for safety and system stability.

Graphical user interface (GUI)

The visual part of the OS, windows, icons, and menus, that lets you interact through clicking and tapping.

Command-line interface (CLI)

A text-based interface where you type commands to control the system with precision and speed.